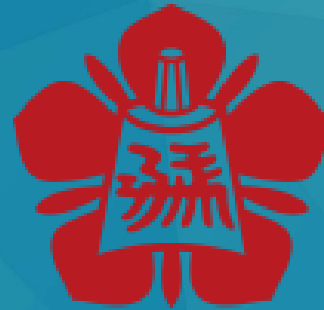


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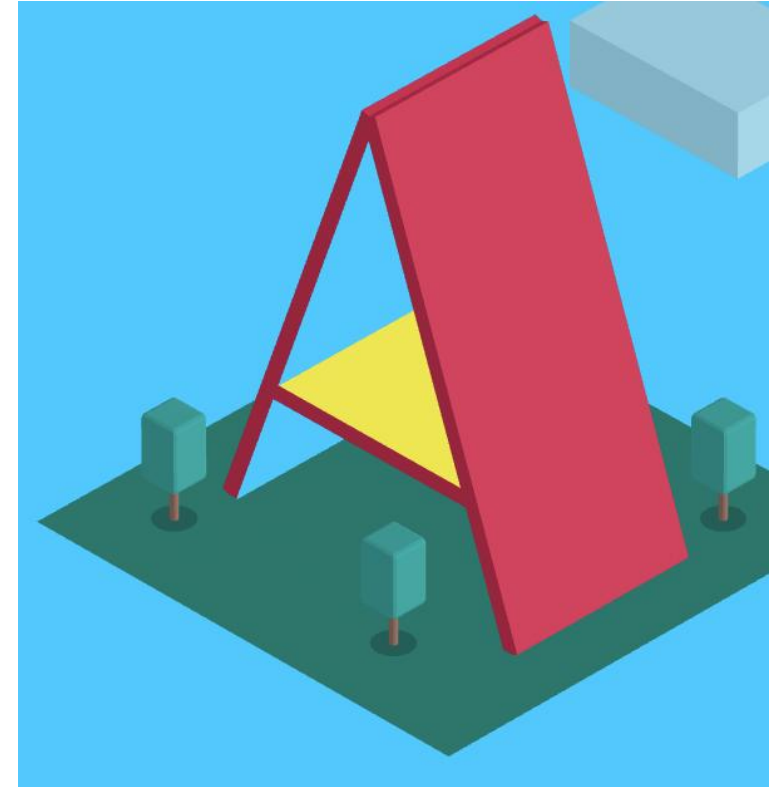
A-Frame WebVR Basic Tutorial

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What is A-Frame?

- [A-Frame](#) is a web framework for building virtual reality (VR) experiences using HTML and JavaScript
- Specifically, it's based on WebGL / [three.js](#)
- Developed by [Mozilla](#) (the company who makes the Firefox browser)



Why WebVR?

- WebVR:
VR content that runs on webpages
- Native VR:
Apps developed by game engines (e.g., Unity or Unreal) that can be directly installed and run on VR headsets (e.g., Oculus, Vive)
- Why WebVR?
 - Easier to develop and less “expensive” (in terms of cost or learning barrier)
 - Can run on any devices (as long as it has a web browser that supports modern webpages)
 - Downsides: less graphics-intensive, limited support for VR controllers, less powerful & versatile for development
 - Strengths: Good for entry level learning (3D, VR, interactions), cross-platform support, conceptual prototyping/sharing

Developing WebVR using A-Frame

- Overall workflow
 - Write the code in a HTML editor
 - Before you see the changes you make, make sure to save the HTML file
 - View the file using a browser
- Choice of HTML editor
 - Basic text editors: Notepad (Windows), TextEdit (Mac), etc.
 - Text editors dedicated for coding: [VS Code](#), [Sublime](#), etc.
- Make sure the filetype of the code is saved as “html”!!

The a-scene tag is for creating A-Frame's scene "entity". All the code for A-Frame goes inside the a-scene opening & closing tags.

```
<html>
  <head>
    <script src="https://aframe.io/releases/1.4.0/aframe.min.js"> </script>
  </head>
  <body>
    <a-scene>
      <a-sky color="#ECECEC"> </a-sky>
    </a-scene>
  </body>
</html>
```

This line adds the A-Frame library.

Feel free to use the latest version.

- The a-sky tag is for creating a "skybox". In 3D editing software, like Unity, skybox is the 3D background that surrounds the scene.
- The color is in hex code. Play around with different colors.

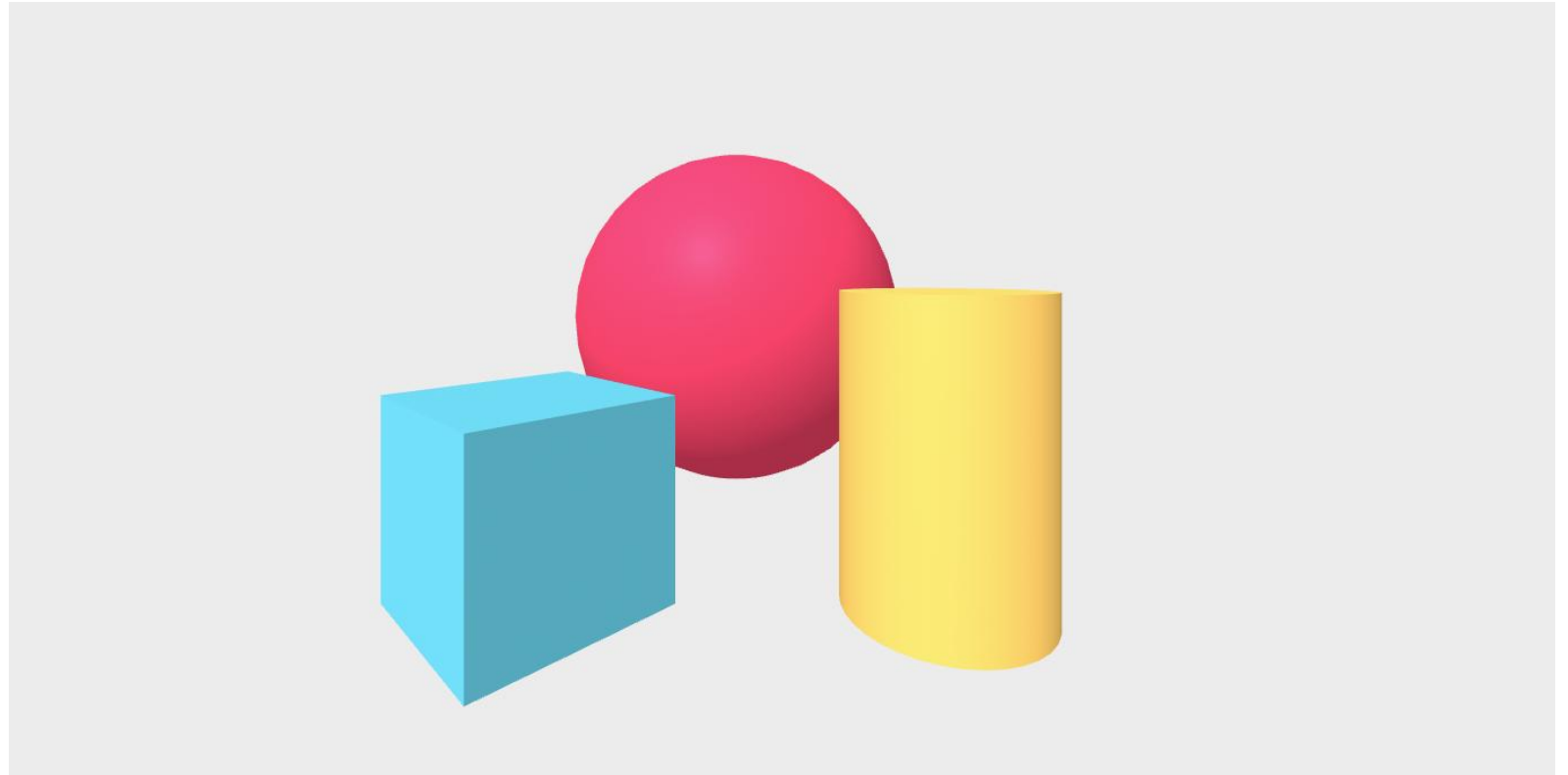
```
<a-sphere position="0 1.25 -5" radius="1.25" color="#EF2D5E"></a-sphere>  
<a-box position="-1 0.5 -3" rotation="0 45 0" width="1" height="1" depth="1" color="#4CC3D9"></a-box>  
<a-cylinder position="1 0.75 -3" radius="0.5" height="1.5" color="#FFC65D">
```

- Adding these tags inside the a-scene tags will render the “primitive” objects. For example, here we are adding a ball, a box, and a cylinder.
- Play round with different values of the attributes: position, rotation, radius, color, etc. And see how they look in your browser.
- The unit is in meter. What does that mean?
- Where is the origin position (0, 0, 0)?

You should have something like this so far.

Use drag
(click+hold+move the mouse) to “look around”.

Use WASD to move.



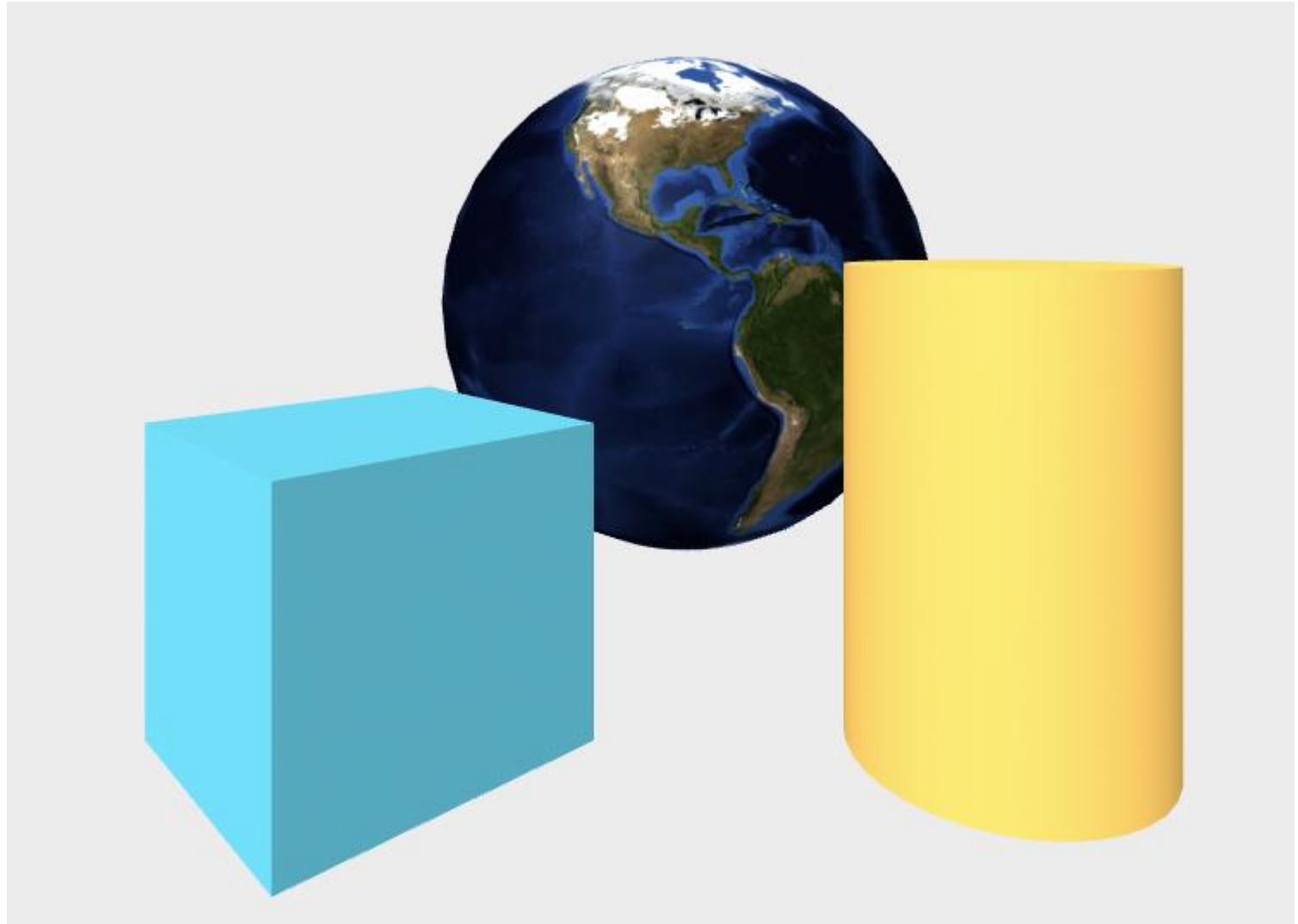
```
<a-sphere position="0 1.25 -5" radius="1.25" src="https://raw.githubusercontent.com/aframevr/sample-assets/master/assets/images/space/earth_atmos_4096.jpg"></a-sphere>
```

- You can specify an image file as the “texture” of the object.
- Here I remove the “color” attribute and added the “src” attribute, and I use an image URL as the texture.
- What happens if we keep both the color and src attributes?

You should have something like this so far.

Use drag
(click+hold+move the mouse) to “look around”.

Use WASD to move.



```
<a-sky src="https://images.unsplash.com/photo-1570284613060-766c33850e00?q=80&w=1000&auto=format&fit=crop&ixlib=rb-4.0.3&ixid=M3wxMjA3fDB8MHxzZWZyY2h8OHx8c3RhcnJ5JTlwc2t5fGVufDB8fDB8fHww"> </a-sky>
```

- In addition to color, you can also use an image as the texture of the skybox.
- Here it is a regular image, which is not ideal for skyboxes. You can refer to this page for better choices of images for creating your skybox:
[A-Frame's a-sky](#)

```
<html>
  <head>
    <script src="https://aframe.io/releases/1.4.0/aframe.min.js"></script>
  </head>
  <body>
    <a-scene>

      <a-sky src="https://images.unsplash.com/photo-1570284613060-766c33850e00?q=80&w=1000&auto=format&fit=crop&ixlib=rb-4.0.3&ixid=M3wxMjA3fDB8MHxzZWFiY2h8OHx8c3RhcnJ5JTlwc2t5fGV" ></a-sky>
      <a-sphere position="0 1.25 -5" radius="1.25" src="https://raw.githubusercontent.com/aframevr/sample-assets/master/assets/images/space/earth_atmos_4096.jpg"></a-sphere>
      <a-box position="-1 0.5 -3" rotation="0 45 0" width="1" height="1" depth="1" color="#4CC3D9"></a-box>
      <a-cylinder position="1 0.75 -3" radius="0.5" height="1.5" color="#FFC65D">

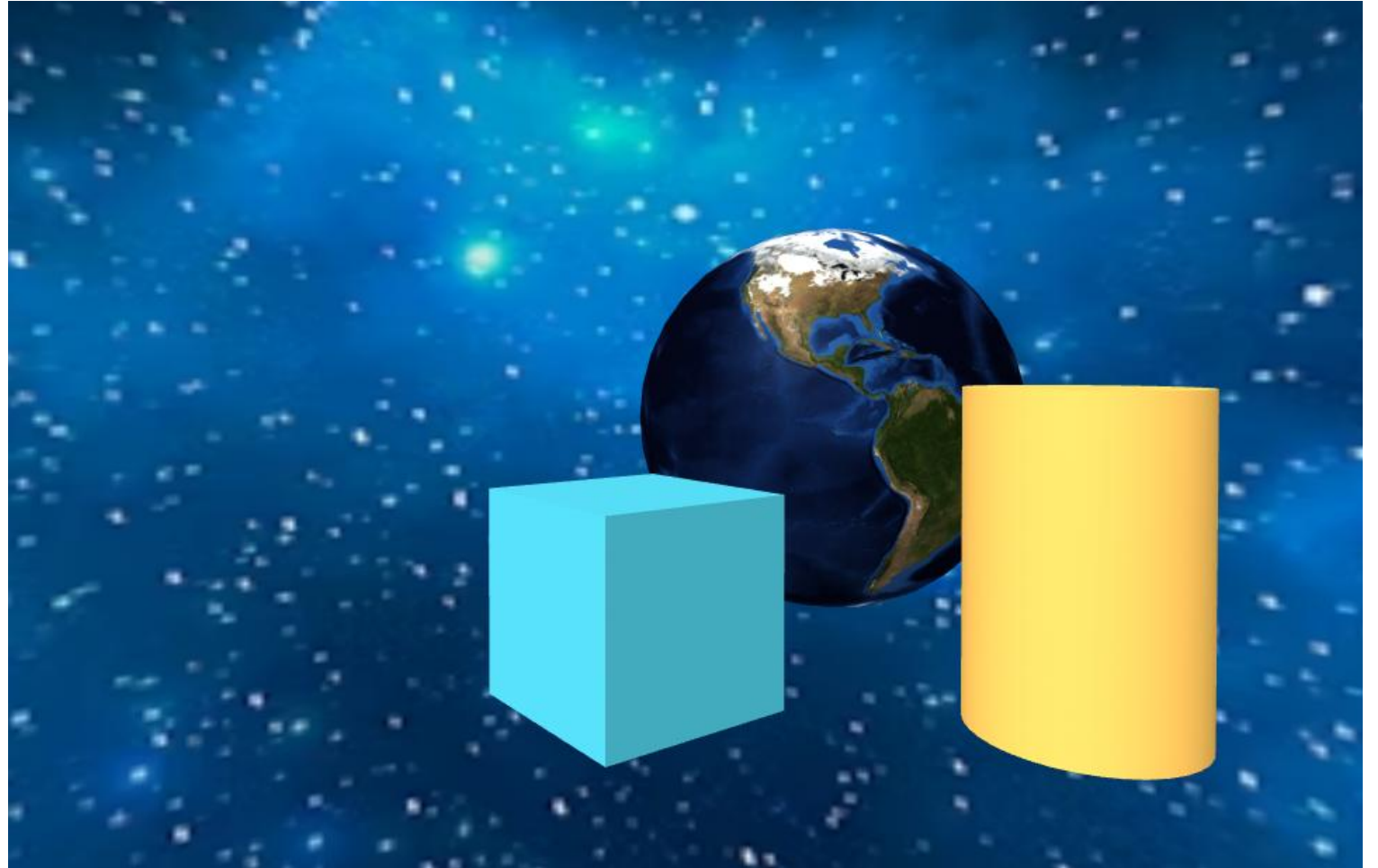
    </a-scene>
  </body>
</html>
```

- Your submitted file's full code should be something like this.
- I purposefully make it as an image so you don't directly copy and paste.
- But overall, if you understand/assemble/modify the codes from previous pages, you should have something like this.
- See the next page for the instructions for this tutorial.

You should have something like this so far.

Use drag
(click+hold+move the mouse) to “look around”.

Use WASD to move.



Other learning resources

- Please use these resources to achieve the Project:
 - [Getting Started – A-Frame \(aframe.io\)](https://aframe.io/docs/1.3.0/getting_started/)
 - [Building a Basic Scene – A-Frame \(aframe.io\)](https://aframe.io/docs/1.3.0/building_a_scene/)
 - [Building up a basic demo with A-Frame - Game development | MDN \(mozilla.org\)](https://developer.mozilla.org/en-US/docs/Web/VR/Building_up_a_basic_demo_with_A-Frame)
 - ...Others you find

Thank You!
Questions?

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